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Melvin Immanuel Gregory

Aspiring AI | ML Engineer | Data Scientist

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OBJECTIVE

Mathematics drives my interest in Artificial Intelligence and Machine Learning, with a strong foundation in Python, deep learning, NLP, LLMs, and model evaluation. I build AI solutions involving RAG, document intelligence, prompt engineering, and data-driven model development, translating mathematical and analytical reasoning into reliable, scalable AI systems.

SKILLS

Programming Python, SQL, R, Java, C++, PL/SQL

ML/DL Scikit-learn, TensorFlow, PyTorch, Keras, Hugging Face Transformers

NLP/LLM NLP, RAG, LangChain, Prompt Engineering, Embeddings, Vector Databases, FastAPI

Document AI EasyOCR, OCR, PDF Processing, Document Extraction, Semantic Search

Data Libraries Pandas, NumPy, Matplotlib, Seaborn

Tools Git/GitHub, Jupyter Notebook, Flask, Power BI, Tableau

Cloud AWS (Basics), Google Colab, Hadoop

Core Concepts GANs, Time Series Forecasting, Regression, Classification, Neural Networks

PROJECTS AND INTERNSHIPS

Agentic RAG-Based Intelligent Analysis and Decision Support for Legal Documents

- Developing an Agentic Retrieval-Augmented Generation (RAG) system for analyzing long-form legal documents and generating context-grounded responses using LLMs.
- Building an end-to-end document intelligence pipeline involving PDF ingestion, text extraction, chunking, embedding generation, vector retrieval, and semantic search.
- Implementing retrieval and prompt-engineering strategies to improve contextual relevance, reduce hallucinations, and provide evidence-grounded legal analysis with source references.
- Exploring agentic workflows for query decomposition, document retrieval, contextual reasoning, and automated decision-support across complex legal documents.
- Using Python, NLP, Hugging Face Transformers, vector databases, LangChain, and LLM-based evaluation to develop and assess the system.

Adversarial Time Series Forecasting using GANs

- Developed a GAN-based framework for time-series imputation and forecasting using Python, TensorFlow, and Pandas.
- Engineered lag, rolling-window, and statistical features to capture temporal dependencies.
- Performed model evaluation using RMSE, MAE, and cross-validation techniques.

Handshake AI — AI Analyst

Jul 2026 – Present

- Analyzed and evaluated AI-generated outputs to assess factual accuracy, reasoning quality, relevance, and adherence to task-specific requirements.
- Applied structured analytical and mathematical reasoning to identify errors, inconsistencies, and failure patterns in AI model responses.
- Contributed to STEM-focused AI projects PLANCK and Lumière, working with OpenAI, Anthropic, and other advanced AI models to improve reasoning and problem-solving capabilities.
- Contributed to improving AI system reliability through systematic error analysis, data quality assessment, and feedback-driven evaluation.

EDUCATION

M.Sc. Data Science

Jun 2025 – Jun 2027

Vellore Institute of Technology (VIT), Chennai

CGPA: 9.02/10

Relevant Coursework: Machine Learning, Deep Learning, Statistical Modeling, Data Mining, Exploratory Data Analysis, Big Data Analytics, Cloud Computing, Business Analytics

B.Sc. Mathematics

Jun 2022 – Jun 2025

Loyola College, Chennai

CGPA: 9.22/10 (TOP 5%)

Relevant Coursework: Linear Algebra, Probability Theory, Statistical Inference, Regression Analysis, Numerical Methods, Design and Algorithms

CERTIFICATIONS & ACHIEVEMENTS

- Stanford Online – Supervised Machine Learning: Regression and Classification
- Stanford Online – Unsupervised Learning, Recommenders & Reinforcement Learning
- Introduction to MongoDB — Simplilearn | Data Structures in C++ — Scaler Topics